

# Use Case: Credit Risk Scoring System

## 1. Goal & Expectation

### Goal

Build a data-driven Credit Risk Scoring System that predicts the likelihood of a loan applicant defaulting, using financial history, behavioral data, and credit bureau information.

### Expectations

- Reduce loan default rates by identifying risky applicants early.
- Increase approval rate for good customers without increasing risk.
- Provide transparent, fair, and explainable credit decisions.
- Ensure compliance with regulatory requirements for fairness and bias mitigation.
- Enable business teams to monitor loan portfolio risk in real-time dashboards.
- Code base availability in Github
- Document the details in a Readme file in github.

## 2. End-to-End Workflow Overview

ETL / Data Engineering → Data Science Modeling → Monitoring → Visualization

## 3. Detailed Use Case Breakdown

### A. Problem Statement

Financial institutions need to assess applicant creditworthiness by analyzing historical and behavioral patterns. Traditional rule-based systems lack flexibility and may introduce bias. The goal is to use machine learning to compute a credit risk score and enable data-informed lending decisions.

### B. Representative Data Sources

#### 1. Loan Application Data

- Applicant demographics
- Employment details
- Declared income
- Loan amount, tenure

#### 2. Transaction History

- Bank account statements
- Spends, inflows/outflows



- Delinquency history
- Outstanding loans

### 3. Credit Bureau Reports

- Credit score
- Past loan performance
- Enquiries
- Credit utilization ratio

## C. ETL / Data Engineering Pipeline

### 1. Data Ingestion

- Batch ingestion from:
  - Core banking systems
  - Bureau APIs
  - Transaction databases
- Streaming ingestion for real-time risk scoring (optional)

### 2. Data Quality Validation

- Missing value check
- Data type & schema validation
- Outlier detection
- Duplicate handling

### 3. Data Cleansing & Transformation

- Imputing missing fields
- Standardizing income & amount formats
- Normalizing categorical values (employment type, loan purpose)

### 4. Feature Engineering

- Financial Ratios: Debt-to-income, credit utilization
- Behavioral Features: Delayed payments, monthly spend stability
- Derived Metrics: Rolling 6-month delinquencies, volatility of account balance
- Bureau Features: Number of past defaults, enquiry rate

### 5. Feature Encoding

- One-hot encoding
- WOE (Weight of Evidence) for explainability
- Target encoding (with leakage prevention)



## 6. Data Storage

- Curated datasets stored in:
  - Data Lake (raw, refined)
  - Data Warehouse (curated, aggregated tables)

## 4. Data Science Modeling

Model

- LightGBM (fast training, good default performance)

Train/Validate split

- 80/20 random split

Model Metrics

- AUC-ROC
- KS statistic

Target:

- $AUC \geq 0.80$
- $KS \geq 30$

Calibration (Optional )

- Use CalibratedClassifierCV if needed

## 5. Dashboard Visualization

Minimal but effective visualization set:

### 1. Portfolio Risk Overview

Histogram of predicted risk scores

### 2. Model Performance

AUC & KS values

### 3. Insight bar chat

### 4. Fairness Check

Comparison of average risk score by gender